

Graph-Guided Reasoning and Refinement in Diffusion Language Models

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Abstract

Diffusion Language Models (DLMs) offer an alternative to autoregressive language generation by iteratively refining partially masked sequences, enabling revision during generation. However, existing correction mechanisms primarily operate at the token or sequence level and do not explicitly represent dependencies between reasoning steps. We propose a graph-guided reasoning framework that represents intermediate reasoning as a dependency graph and uses this structure to guide denoising and localized error correction. By selectively re-masking only the reasoning subgraph associated with an inconsistency, our approach aims to improve reasoning consistency while reducing unnecessary regeneration, and potentially improve computational efficiency. We plan to evaluate the framework on multi-step reasoning benchmarks and compare it against existing diffusion-based reasoning and correction methods.

1 Motivation

Large Language Models (LLMs) have demonstrated strong performance on a wide range of reasoning tasks. However, most LLMs are autoregressive, generating text sequentially from left to right (Li et al., 2026). While sequential generation is simple and effective, once a token has been generated, it cannot be revised. Consequently, errors made in the reasoning process propagate through subsequent steps and negatively impact the final output (Yu et al., 2026). Although techniques such as self-consistency, verification, and post-hoc analysis can mitigate these issues, they often require regenerating large portions of the reasoning process and do not provide a straightforward mechanism for revising specific reasoning steps.

Diffusion Language Models (DLMs) have recently emerged as an alternative in natural language processing (NLP) (Li et al., 2026). Rather than pro-

ducing text token-by-token, DLMs iteratively denoise a partially masked sequence until a complete output is obtained. This iterative process allows for refinement and correction during generation. As a result, DLMs provide a potential framework for error correction and reasoning refinement not possible in autoregressive models (Ye et al., 2024). Recent work has explored diffusion-based reasoning, Chain-of-Thought (CoT) generation, and self-correction mechanisms, demonstrating that refinement during generation can improve reasoning quality (Ye et al., 2024; Huang et al., 2025; Schiff et al., 2026). Several approaches have introduced structure into DLM generation through logical constraints, causal concepts, and graph-based representation (Aman, 2026; Han et al., 2025).

Despite these advances, existing structured diffusion approaches primarily use graphs for reasoning representation, search, or generation guidance (Besta et al., 2024; Han et al., 2025). However, existing correction methods primarily operate through remasking and iterative refinement at the token or sequence level, treating reasoning as unstructured text (Huang et al., 2025; Schiff et al., 2026). As a result, when inconsistencies are detected in the reasoning space, the model lacks an explicit representation of how reasoning steps depend on each other, making it difficult to determine and fix which portions of the reasoning process are responsible for the error and which can be safely preserved. Additionally, correction may require revisiting large regions of otherwise valid reasoning, leading to unnecessary computation and potentially introducing new errors.

We argue that reasoning structure can play an active role in the error-correction process. Many reasoning tasks naturally form dependency graphs in which future conclusions rely on earlier steps while others remain independent. We therefore propose a graph-guided diffusion framework that represents reasoning as a dependency graph. Un-

like previous graph-based reasoning methods that primarily use graphs for reasoning representation, search, or generation control, our approach uses the graph as a mechanism for refinement or correction during diffusion. We hypothesize that graph-guided refinement can improve both efficiency and reliability in diffusion-based reasoning. By using the dependency structure to identify and repair only the minimal reasoning subgraph related to the contradiction, a diffusion language model may avoid unnecessary regeneration while also preserving correct intermediate conclusions. We aim to evaluate whether graph-guided denoising and subgraph-level correction provide improvements in reasoning consistency, computational efficiency, and overall accuracy in DLMs over existing diffusion-based reasoning and correction methods.

2 Proposed Methodology

We propose a graph-guided reasoning framework for Diffusion Language Models (DLMs). Given a reasoning task, the model first generates an intermediate reasoning trace, which is represented as a dependency graph where nodes correspond to reasoning steps or claims and directed edges represent the dependency relationships between them. The graph may be constructed from intermediate reasoning steps generated by the model, with dependencies inferred from explicit references between steps or extracted using a lightweight parsing procedure.

The dependency graph is used in two ways. First, we investigate dependency-aware denoising schedules, where nodes whose prerequisites have been resolved are prioritized for refinement. Independent branches of the graph may be denoised in parallel, while dependent nodes are refined only after their supporting reasoning has stabilized. Second, we investigate localized error correction. We introduce an inconsistency detection mechanism that monitors generated reasoning for inconsistencies. When a contradiction is detected, the framework identifies the smallest dependency subgraph associated with the conflict and selectively remasks only the affected nodes. The resulting subgraph is then re-denoised while preserving the remainder of the reasoning trace. Unlike existing correction methods that often operate at the token or sequence level, this approach performs correction at the level of reasoning structure. We hypothesize that such targeted refinement will be more computationally efficient,

as it avoids regenerating large portions of otherwise correct reasoning and focuses computation only on the region responsible for the inconsistency.

To evaluate this framework, we plan to conduct experiments on multi-step reasoning benchmarks, such as GSM8K, and other mathematical reasoning tasks. We will compare our framework to a baseline DLM along with other existing self-correcting and reasoning DLM frameworks. Evaluation will be primarily focused on reasoning accuracy, reasoning consistency, and computational efficiency.

References

- Shaik Aman. 2026. [Logicdiff: Logic-guided denoising improves zero-shot reasoning in masked diffusion language models](#). *Preprint*, arXiv:2603.26771.
- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoefer. 2024. [Graph of thoughts: Solving elaborate problems with large language models](#). *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16):17682–17690.
- Kairong Han, Nuanqiao Shan, Ziyu Zhao, Zijing Hu, Xinpeng Dong, Junjian Ye, Lujia Pan, Fei Wu, and Kun Kuang. 2025. [C²dml: Causal concept-guided diffusion large language models](#). *Preprint*, arXiv:2511.22146.
- Zemin Huang, Yuhang Wang, Zhiyang Chen, and Guojun Qi. 2025. [Don’t settle too early: Self-reflective remasking for diffusion language models](#). *Preprint*, arXiv:2509.23653.
- Tianyi Li, Mingda Chen, Bowei Guo, and Zhiqiang Shen. 2026. [A survey on diffusion language models](#). *Preprint*, arXiv:2508.10875.
- Yair Schiff, Omer Belhasin, Roy Uziel, Guanghan Wang, Marianne Arriola, Gilad Turok, Ran Zilberstein, Michael Elad, and Volodymyr Kuleshov. 2026. [Learn from your mistakes: Self-correcting masked diffusion models](#). *Preprint*, arXiv:2602.11590.
- Jiacheng Ye, Shansan Gong, Liheng Chen, Lin Zheng, Jiahui Gao, Han Shi, Chuan Wu, Xin Jiang, Zhenguo Li, Wei Bi, and Lingpeng Kong. 2024. [Diffusion of thoughts: Chain-of-thought reasoning in diffusion language models](#). *Preprint*, arXiv:2402.07754.
- Longxuan Yu, Yu Fu, Shaorong Zhang, Hui Liu, Mukund Varma T, Greg Ver Steeg, and Yue Dong. 2026. [Thinking out of order: When output order stops reflecting reasoning order in diffusion language models](#). *Preprint*, arXiv:2601.22035.